

Building Socially Conscious Systems: A Research Vision for Frugal AI, Space Connectivity and Culturally Embedded Data Governance in Frontier Markets

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Abstract. More than two billion people remain offline. Hundreds of millions more are connected only in the thinnest sense. They can send a message, but they cannot be seen, heard, or paid on their own terms. At the same time, a large share of the world’s linguistic and cultural heritage is at risk of disappearing within a generation. Often this is the same heritage held by communities that connectivity infrastructure has historically bypassed. This paper sets out a research vision for building *socially conscious systems*. These are systems that close both gaps together, by design. They combine frugal artificial intelligence, space based connectivity and compute, and privacy preserving data architectures. Cultural and religious norms are embedded directly into how content is captured, shared and monetised, rather than added as an afterthought. We argue that these elements are not independent research tracks. They are a single system. Connectivity without cultural agency reproduces extraction. Cultural preservation without economic participation is unsustainable. We ground the vision in early evidence from the Saving Voices Initiative, which is building frugal, community governed speech technology with tribal communities in India. We give concrete examples of how specific cultural and religious practices should shape technical policy. We then set out a research agenda spanning frontier and emerging markets more broadly, from Sub-Saharan Africa to South and Southeast Asia to Latin America. We argue that these markets are best understood not as places technology eventually reaches, but as places where genuinely new technical and economic models are likely to be built first.

Keywords: frugal AI, edge computing, satellite connectivity, orbital compute, data sovereignty, privacy by design, cultural and religious norms, digital inclusion, frontier markets, creative economy

1 Introduction: The Unserved and the Unheard

Two distinct but deeply related gaps define the digital exclusion of the twenty first century. The first is a connectivity gap. As of 2025, roughly 2.2 billion people remain offline. Even among those who are connected, quality of access diverges sharply by geography and income. A typical user in a high income country generates nearly eight times more mobile data than a user in a low income country (ITU, 2025). The second, less discussed gap is a voice gap. Even where connectivity arrives, the platforms, models and marketplaces built on top of it were rarely designed with the world’s smaller languages, oral traditions and cultural economies in mind. Language technology in particular has developed almost entirely around a small number of high resource languages. This leaves the majority of the world’s roughly 7,000 languages, of which around 40% are considered endangered (Moseley, 2010), effectively invisible to the tools reshaping how information and culture circulate.

These two gaps compound each other. A community that is unconnected cannot participate in the digital economy at all. A community that is connected but unheard is offered participation on someone else’s terms. Its language may be one a voice assistant cannot transcribe. Its music may be one a streaming platform cannot recommend. Its sacred material may be indistinguishable, to a data hungry model, from public content. Historically, when connectivity and data collection have arrived in underserved regions without accompanying protections, the result has often resembled what Couldry and Mejias (2019) term *data colonialism*: the appropriation of human life and

knowledge as a raw input for value created and captured elsewhere. Bird (2020) makes the same point specifically for language technology. He argues that treating Indigenous languages as “data ready for the taking” risks reenacting the very dynamics that endangered those languages in the first place.

This paper argues that three converging technologies now make it possible to close both gaps at once: frugal artificial intelligence; low Earth orbit (LEO) satellite connectivity and compute; and privacy by design data architectures that embed cultural and religious norms directly into system behaviour. Crucially, they can do so on terms set by the communities themselves, rather than by the infrastructure providers who reach them. We call the resulting design target a *socially conscious system*. In such a system, cultural sensitivity, consent and fair economic return are not compliance features bolted onto a working product. They are constraints that shape the system’s architecture from the outset, in the same way that latency or bandwidth would. We treat this as a single research vision rather than separate technical projects, because each element is largely inert without the others. Connectivity without cultural embedding accelerates extraction. Frugal AI without new monetisation models preserves culture without sustaining the people who hold it. Monetisation without genuine community control simply digitises the same asymmetries that have long characterised the relationship between frontier markets and the technology industry.

2 The Opportunity in Frontier and Emerging Markets

The framing of “frontier markets” matters here. These economies span Sub-Saharan Africa, South and Southeast Asia, Latin America and elsewhere. They are conventionally treated in technology strategy as destinations: places existing products eventually reach once markets in wealthier economies are saturated. Frugal innovation research suggests the opposite framing is often more accurate, and more useful. Radjou, Prabhu and Ahuja (2012) document how resource constrained environments routinely produce more original, not merely cheaper, solutions than resource rich ones. Genuine constraint forces a reexamination of what a technology is actually for, rather than an incremental optimisation of an existing design. Radjou and Prabhu (2015) extend this into a broader argument: frugality, doing more with less, is becoming a source of competitive advantage that flows from emerging markets outward, rather than the reverse. Prabhu, Narayan and Sud (2026) develop this argument specifically for India, showing how frugal, high tech innovation across drones, electric vehicles, fintech, artificial intelligence and space missions is turning resource constraint into a repeatable, exportable capability, rather than a one off workaround. The technical pillars set out in Section 3 are best read as one instance of that same pattern, applied to oral and musical heritage.

Mobile financial services provide the clearest existing precedent. Jack and Suri (2014) show that Kenya’s M-PESA mobile money system measurably improved households’ ability to absorb income shocks. It did this not by replicating a Western banking product on a phone, but by building a genuinely new financial instrument suited to an economy with low bank penetration and high mobile phone penetration. The lesson generalises. Frontier markets do not need a smaller version of infrastructure built elsewhere. They need infrastructure designed from their actual constraints outward. When that happens, the resulting model is often exportable back to wealthier markets, not just imported into poorer ones.

The same logic applies directly to cultural and creative economies. Sub-Saharan Africa’s music industry grew faster than any region in the world in 2024, up 24% in a single year (AfriLabs, 2024). The United Nations Conference on Trade and Development projects Africa’s creative goods and services exports could be worth in the order of \$200 billion a year by 2030 (UNCTAD, 2024). Yet the platforms, distribution infrastructure and ownership structures capturing this growth remain overwhelmingly built and controlled outside the markets producing the value. This is precisely the pattern the research vision in this paper is designed to interrupt. The aim is not to deny frontier markets access to global platforms. It is to ensure the underlying technology, connectivity, models and data architecture, is built so that value capture and cultural control can

remain with the communities generating it.

3 Technical Pillars

We organise the technical research agenda around four elements. Each is grounded in an existing, active research literature. The contribution of this vision is to argue that they must be designed together, as one socially conscious system.

3.1 Frugal AI

Frugal AI is frequently, and incompletely, equated with TinyML: the practice of building machine learning models small and efficient enough to run inference directly on low power hardware (Abadade et al., 2023). Model level efficiency is one important technique. But the Frugal AI Hub at Cambridge Judge Business School, drawing on the frugal innovation tradition (Radjou, Prabhu and Ahuja, 2012), defines frugal AI more broadly. It is a design philosophy organised around four interconnected pillars: resource efficiency, minimising consumption of energy, computational power, data and financial capital across the AI lifecycle; sustainability, reducing the carbon footprint of training and inference infrastructure; accessibility and inclusion, democratising access to AI capability for organisations and communities that cannot afford state of the art compute; and impact and scalability, allowing a system to scale its social and economic benefit without a proportional increase in infrastructure investment (Frugal AI Hub, 2025). As the White Paper’s foreword puts it, “frugal innovation is not just about cost cutting; it is about reimagining processes, products, and systems to create value under constraints” (Prabhu, in Frugal AI Hub, 2025). This is the sense in which we use the term throughout this paper. Frugal AI is an architectural and economic philosophy. Model compression and edge deployment are important instruments within it, not the whole definition.

For a community documenting an oral tradition or an endangered language, the resource efficiency and accessibility pillars matter directly. They determine whether preservation requires reliable high bandwidth internet access, which the community may not have, or can happen directly on a low cost device in the room where the recording is made. The practical toolkit within this pillar includes model compression, quantisation (reducing numerical precision from, for example, 32-bit to 8-bit representations), pruning of redundant parameters, and knowledge distillation, training a small “student” model to reproduce the behaviour of a much larger one. Together, these allow a general purpose audio or speech foundation model, trained on large, generic datasets, to be fine tuned and compressed for a specific, low resource task at a fraction of the original computational cost (Han and Siebert, 2020). In earlier work, we argue that fine tuning alone is not enough for languages this different from the high resource languages such models were built around (Sathiaselam, 2026). We propose byte and morpheme aware tokenisation for languages with no standardised orthography, typologically informed self supervised pretraining, and parameter efficient fine tuning (PEFT) methods that adapt a small number of parameters rather than the whole model, so that a handful of hours of recorded speech is enough to produce a usable system. This matters for the vision proposed here. A community need not generate a large labelled dataset to see benefit. A general model already carries substantial useful structure, and the community’s own recordings become fine tuning material, rather than a training set built from scratch. The impact and scalability pillar, less discussed in the technical literature, is equally central to this vision. A frugal system’s ultimate test is not benchmark accuracy. It is whether the system can scale from one community to thousands, without requiring proportionally more infrastructure investment from any of them.

A crucial extension of this pillar is still underexplored in the frugal AI literature: embedding cultural sensitivity directly into model behaviour, rather than treating it as a post hoc filter. Bird (2020) argues that speech and language technologists need to move beyond treating language communities purely as data sources. Instead, they should adopt participatory design, in which the community shapes what the technology is built to recognise and protect, not only what it is

built to transcribe. A concrete research question follows: can a frugal model be trained not only to recognise a voice or a song, but to recognise, at the point of capture, the category of material it belongs to, sacred, ceremony bound, freely shareable, so that appropriate handling begins before the recording ever leaves the device. Section 3.3 develops this question, and the cultural and religious reasoning behind it, in detail.

3.2 Space Based Connectivity and Compute

Terrestrial network infrastructure has consistently reached frontier and rural markets last. The capital cost of laying fibre or building cell towers does not scale down gracefully with population density. Alternative Network Deployments research documents a range of models built to close this gap: community networks, shared infrastructure networks, and low cost wireless internet service providers, all developed specifically to serve populations that mainstream commercial deployment does not reach (Saldana, Arcia-Moret, Braem, Pietrosevoli, Sathiaselan and Zennaro, 2016). A systematic review of 21 community network deployments across multiple continents finds that sustainability, not initial deployment, is the binding constraint on these models (Gwaka, Haseki and Yoo, 2023). This is itself a signal that connectivity alone, without an accompanying economic model, is an incomplete intervention, a theme we return to below.

The more recent shift toward large low Earth orbit satellite constellations changes this calculus. LEO satellites are cheap to manufacture relative to earlier monolithic satellites, and can be deployed in large numbers. They therefore offer a path to connectivity that does not require terrestrial capital investment proportional to the population served. This is precisely the constraint that has historically excluded remote and low density communities. Beyond connectivity, a newer body of work treats the satellites themselves as a computing layer. Denby and Lucia (2020) introduce *orbital edge computing*: performing computation, including machine learning inference, directly onboard nanosatellite constellations, rather than downlinking raw data to be processed on the ground. This is motivated by the observation that a satellite typically generates far more data than its limited downlink bandwidth can transmit in a single pass.

For the vision set out in this paper, orbital edge computing suggests a genuinely novel architectural possibility. It is not simply “satellite internet for remote villages.” A frugal model trained to recognise and tag culturally sensitive material could, in principle, run part of its aggregation step on an orbital compute node, rather than a server controlled by a single national government or corporation. This arrangement has a useful property: no single terrestrial jurisdiction holds unilateral authority over the aggregation layer. Whether this property is meaningful in practice is an open empirical question. It depends on latency, cost, legal jurisdiction over data in orbit, and genuine community trust, and it is a priority for the research agenda in Section 6.

3.3 Privacy by Design and Embedding Cultural Principles into AI

Where the previous two pillars concern how data is captured and moved, this pillar concerns who controls it once it exists. The starting point is a body of work in Indigenous data governance that predates, and substantially anticipates, current mainstream debates about AI and privacy. The CARE Principles for Indigenous Data Governance, Collective Benefit, Authority to Control, Responsibility, and Ethics, were developed specifically to complement data centric standards such as the FAIR principles (Findable, Accessible, Interoperable, Reusable). They add a people and purpose oriented framework that recognises data governance as a question of self determination, not merely of technical access (Carroll et al., 2020). Christen (2012) documents one of the earliest working implementations of this idea: the Mukurtu digital archive platform, built with the Warumungu community in Australia. It allows cultural protocols, who may view what, under what conditions, based on community status, gender or ceremonial role, to be encoded directly into a content management system, rather than left to an external institution’s discretion.

The technical contribution this vision adds to that lineage is treating access, sharing and monetisation rules as a single machine readable policy object. This object is attached to each

recording and evaluated automatically at the point of query, rather than read and enforced by human curators. Federated learning provides the mechanism by which a shared model can improve across many such community held stores without centralising the underlying data. McMahan, Moore, Ramage, Hampson and Agüera y Arcas (2017) show that a shared model can be trained by aggregating locally computed updates from many devices, leaving the raw, privacy sensitive data distributed rather than pooled. Combining this architecture with CARE aligned governance is, to our knowledge, largely unexplored in the frugal AI literature. It is one of the more significant open research questions we identify in Section 6: can a federated aggregation step be designed so that what crosses the boundary of a community’s data store is provably insufficient to reconstruct or reidentify a sacred or restricted recording, rather than merely policy restricted from doing so.

How specific cultural and religious practices should shape system behaviour

A generic “privacy setting” is not adequate here. It cannot represent the range of ways cultural and religious practice actually govern access to oral and musical material. The following examples are drawn from documented practice across several traditions. They illustrate the kind of distinctions a socially conscious system needs to represent natively, in its access, sharing and monetisation logic, rather than as an exception handled manually after the fact.

- **Gender restricted material.** Christen (2012) documents how, for Central Australian Aboriginal communities working with the Mukurtu platform, some ceremonial content is restricted by the gender of the viewer. Material associated with men’s ceremonial business may be withheld from women users, and vice versa. This distinction is encoded directly into the archive’s access profiles, rather than left to a viewer’s discretion. For a policy as code architecture, this means a recording’s policy object must carry a viewer attribute condition, not merely a binary public or restricted flag. That condition must be evaluated before any excerpt, waveform, or embedding is returned, not only before the full recording is released.
- **Restrictions tied to initiation, status or seasonal authority.** Many oral traditions restrict material not by who a person is permanently, but by what they have been through, or when it is. A chant or story may only be appropriately heard by someone who has completed a particular stage of initiation. It may only be performed or played during a specific season or ceremonial period. A system built around a single, permanent access tier cannot represent this. The policy object needs a temporal or conditional dimension: for example, a rule that unlocks a recording only during a defined ceremonial window each year, or that requires an attestation of status from a recognised community authority before access is granted, rather than a one time permission check.
- **Material connected to deceased persons.** Several Indigenous protocols, again documented in the Mukurtu literature (Christen, 2012), restrict the circulation of a deceased person’s name, image or voice. This may apply for a mourning period, or require family consent before such material is shared again. A frugal capture pipeline that automatically tags a voice recording with a speaker identity should therefore also support an automatic suspension trigger. If a community reports that a recorded speaker has died, the system should be able to move that speaker’s material into a restricted state pending renewed authorisation, rather than requiring a manual takedown request to an external platform.
- **Sama and devotional listening in Sufi practice.** In several Sufi devotional traditions, certain forms of sung dhikr and qawwali are understood within a framework of *sama*: sacred listening intended to produce a specific spiritual state in a specific communal setting, rather than music for general entertainment consumption. A recording of such material licensed for a commercial soundtrack, without any accompanying context, can be experienced, from within that tradition, as a category error, not merely a commercial one. A socially conscious monetisation layer should therefore attach a *context of use* condition to a licence. It could permit streaming or broadcast use, but restrict synchronisation with unrelated commercial

content, a distinction current music licensing marketplaces do not typically represent at all.

- **Māori kaitiakitanga and the Te Hiku Media Kaitiakitanga License.** Te Hiku Media, the Far North New Zealand iwi media organisation building speech technology for te reo Māori, developed the Kaitiakitanga (guardianship) License. It did so because standard open data and creative commons licences had no mechanism to represent Māori data as something held in trust, rather than owned outright, or to require that any benefit derived from the data return to the people it came from (Te Hiku Media, 2021). Translated into policy object terms, this is a monetisation condition, not merely an access condition. A licence to use the data can be granted, but only paired with an enforceable benefit sharing obligation back to the originating community, evaluated and settled automatically at the point of commercial use, rather than negotiated separately after the fact.

These examples share a common technical implication. A workable policy object cannot be a single access tier chosen from a short, fixed list. It needs to represent conditions on the viewer (status, gender, initiation level), conditions on time (season, ceremonial calendar, mourning period), and conditions on context and downstream use (synchronisation restrictions, benefit sharing obligations). These conditions must be evaluated together at the point a request is made. Designing a schema expressive enough to represent this range of practice, without collapsing distinct cultural logics into one generic permissions model, is one of the central open problems this research programme sets out to address.

3.4 New Monetisation Models and Markets

The final pillar addresses sustainability directly. A system that protects culture but generates no return for the people sustaining it will not persist, however well designed its technical safeguards. Existing proposals for bottom up, community governed data institutions offer a starting structural vocabulary, rather than a finished blueprint. Delacroix and Lawrence (2019) propose *data trusts*: legal structures in which a trustee, bound by fiduciary duty, exercises data rights on behalf of a defined group of beneficiaries. These are explicitly designed as a plural alternative to a single, one size fits all data governance regime, and they matter here because they separate two questions current platforms usually collapse into one: who technically holds a recording, and who has the authority to decide what happens to it. Scholz (2016) develops a parallel argument for *platform cooperativism*. Here, the ownership and governance of a digital platform itself, not only the data flowing through it, is held cooperatively by the people who generate its value, as a structural alternative to extractive platform capitalism, drawing on a much older tradition of worker and agricultural cooperatives applied to digital infrastructure.

Translated into the architecture proposed here, this suggests several concrete, researchable monetisation mechanisms, each of which raises its own design and governance questions.

The first is automated micro licensing. A licensee's request for a recording would be evaluated against the community's own policy object at query time, with a price and, where applicable, a royalty split proposed automatically rather than negotiated case by case. Collecting societies for recorded music, such as PRS for Music or ASCAP, already perform a version of this at scale for conventional commercial recordings, tracking use and distributing royalties without a bespoke negotiation for every play. The contribution proposed here is to push that same logic down to the level of an individual, community held recording, and to let the community, not a national collecting society, set the applicable tier and price. Automation of this kind may reduce transaction costs for routine, low value uses, but it does not by itself resolve who holds valid authority to license a given recording, how ownership disputes within or between communities are settled, or how the resulting payment is reliably identified, verified and delivered across borders. These require payment infrastructure, identity verification and a dispute resolution process that sit alongside the technical system, not inside it.

The second is data cooperatives, in which multiple communities or lineages pool bargaining power

over licensing terms, echoing the collective action logic of platform cooperativism and the historical precedent of agricultural marketing cooperatives, where individually small producers gain leverage by negotiating collectively rather than one at a time. A cooperative structure could plausibly negotiate better commercial terms and share the cost of legal and technical infrastructure across its members. It also concentrates a new kind of risk: a cooperative is only as representative as its governance, and it may be captured by whichever member community is best organised, most fluent in licensing negotiations, or most connected to the researchers and platforms on the other side of the table. Rotating representation, independent audit of how revenue is distributed among members, and a low friction exit option for any community that wants to license independently are the kind of safeguards this design question would need to work through, rather than assumptions the architecture can make for it.

The third is standing fees, paid by platforms that fine tune shared models on federated embeddings without ever accessing raw recordings. This has a directional precedent: stock media and music platforms have already begun negotiating blanket licences for the use of their catalogues in training generative AI systems, which suggests that a metered, ongoing fee for model access, distinct from a one off transaction, is commercially workable in adjacent markets. Applying it here would require a reliable way to measure how much a given community’s material actually contributed to a shared model’s performance, and an audit mechanism both sides trust, neither of which the architecture currently specifies.

The fourth is an explicit noncommercial tier, for material a community designates as sacred or otherwise outside any commercial frame, sustained by grant and philanthropic funding rather than market transactions. This is not a novel funding model in itself, endangered language documentation has long relied on exactly this kind of funding, for example through national research councils and dedicated documentation programmes, but folding it into the same policy object architecture means a recording’s commercial and noncommercial status can be set, and changed, by the community using the same mechanism that governs everything else, rather than requiring a separate application process each time. One risk is worth naming directly: if the noncommercial tier is also the only tier with a guaranteed funding stream, that creates a perverse incentive to classify more material as sacred than a community would otherwise choose to, simply because that is where reliable funding sits. Any implementation would need to monitor for, and correct against, that distortion.

Jack and Suri’s (2014) finding, that a well designed financial instrument can measurably improve household resilience in a frontier market, is a useful benchmark for what success looks like here. It is not simply that a new payment rail exists, or that it is technically elegant. It is that the rail demonstrably changes economic outcomes for the people using it, at a transaction volume and cost structure that makes it viable rather than a subsidised demonstration. The economic viability and distributive effects of each of these four mechanisms, individually and combined, are not established by this architecture; they are hypotheses to be piloted and evaluated against that benchmark, not claims this paper makes in advance of that evidence.

4 An Integrated Architecture

The four elements converge on a single working architecture. A community captures oral or musical material using low power, frugal AI models running on inexpensive local hardware. A community champion, following participatory design principles (Bird, 2020), tags material by cultural sensitivity at the point of capture. The recording, together with its access, sharing and monetisation policy, is stored in a community held, privacy by design data store, rather than a centralised platform. Where terrestrial connectivity does not reach the community, space based connectivity and, potentially, orbital compute carry queries and federated model updates, without requiring raw data to leave the store. Access requests, whether from a researcher, a streaming platform, or a film studio, are evaluated automatically against the recording’s policy object, with any payment routed algorithmically to the community. Each stage depends on the

others. Frugal capture is undermined if the resulting recording is then centralised without consent. Space connectivity is of limited use if what it carries is raw, unprotected data. Monetisation is not meaningful unless the community genuinely controls the terms under which their material is queried in the first place.

5 Illustrative Grounding: Early Evidence

This architecture is not wholly speculative. Elements of it are already operating, in early form, through the Saving Voices Initiative (www.savingvoices.org), which is building frugal, offline, community governed speech technology for languages at risk of disappearing. Its first case is the Soliga tribe of the Biligiri Rangaswamy Hills in Karnataka, India, whose language is exclusively oral and has no widely adopted standardised script (Banerjee Belur et al., 2026). We set out the technical design principles behind this work in detail elsewhere: a rethinking of tokenisation, pretraining and evaluation specifically for underrepresented and indigenous languages, rather than an incremental adaptation of pipelines built for English (Sathiaseelan, 2026). This work follows the participatory pattern the literature recommends. Community members are trained as documentation partners, rather than treated purely as data sources. This directly addresses the concern Bird (2020) raises about speech technology reenacting the causes of language endangerment. We show that a community governed, federated speech recognition architecture can approach the accuracy of a centralised baseline, while running entirely on low cost, intermittently connected devices (Banerjee Belur et al., 2026). It demonstrates that a frugal, on device model can be built and deployed with a small, community sourced corpus, rather than the vast labelled datasets typical of mainstream speech technology. The resulting tool can also function without reliable connectivity, the same constraint the space based connectivity pillar is designed to relax further. What this early work has not yet incorporated at scale is the privacy by design policy layer, the culturally conditioned access logic set out in Section 3.3, and the automated monetisation architecture described above. Extending the Saving Voices architecture in that direction, across additional musical and oral heritage domains, is the next stage of this research programme, and the context in which the present vision paper was developed.

6 Research Agenda and Open Questions

We close by setting out the open questions we consider central to this programme. They are organised by pillar, though several cut across all four, and we deliberately state them as questions rather than as claims the architecture already answers. Progress on this programme should be judged by how well it answers these questions empirically, not by how elegantly the architecture is described.

Frugal AI

The central uncertainty here is not whether a frugal model can be built, but what it can reliably be trusted to do on affordable hardware, in the specific conditions this programme cares about.

- Can a frugal model be trained to recognise the category of cultural sensitivity a recording belongs to at the point of capture, rather than only its acoustic content, and how does its reliability degrade as model size and power budget shrink toward the low end of affordable hardware?
- How much labelled data does this require, relative to conventional transcription tasks, and does that requirement scale differently across languages with very different phonological and morphological structure?
- What is the practical floor on model size and power consumption for a device that must run capture, sensitivity tagging and embedding generation in a single pass, and where does that floor force a trade off between accuracy and battery life or unit cost in the field?

- What evaluation metrics are appropriate for a culturally embedded model, given that a false negative on a sensitivity classification (treating restricted material as shareable) carries a different, asymmetric cost to a false positive, and standard accuracy metrics do not capture that asymmetry?

Space Based Connectivity and Compute

The open questions here concern whether the architectural possibilities this pillar raises hold up once tested against real deployment conditions, rather than against the ideal case.

- Does performing federated aggregation on an orbital compute node offer a meaningful jurisdictional or trust advantage over terrestrial aggregation, or does it merely relocate the same governance question into a harder to audit environment?
- What is the realistic latency and reliability profile for a licensing query that must complete over an intermittent low Earth orbit pass, and for which use cases, real time synchronisation versus batched daily settlement, is this acceptable?
- How does service reliability vary with terrain, weather and ground station placement across the specific frontier and emerging markets this programme targets, and does that variation create new inequities between communities that happen to have good satellite visibility and those that do not?
- What happens to a community's access, and to any pending licensing transaction, if a single commercial constellation operator changes pricing, coverage or terms of service, and does the architecture need to be portable across operators to avoid recreating a single point of dependency?

Privacy by Design and Cultural Embedding

The open questions here concern whether the technical and governance layers can actually deliver on the protections this vision claims for them, rather than only gesturing toward established frameworks.

- Can a federated learning system be designed so that the information leaving a community's data store is provably insufficient to reconstruct a restricted recording, rather than relying on policy alone to prevent misuse, and what threat model would such a proof need to assume?
- How should a system built on CARE principles (Carroll et al., 2020) handle contested authority, where a recording's ownership or sensitivity is disputed within or between communities, rather than assuming a single, unambiguous decision maker exists?
- What mechanism allows a community to revise, suspend or override an automated policy decision after the fact, and how quickly can that override propagate to every party that has already queried or cached the material?
- Whose responsibility is it, technically and institutionally, to verify that the person or body issuing an instruction on a community's data store genuinely holds the authority they claim, and what happens when that authority itself is not settled?

New Monetisation Models

The open questions here are primarily economic and institutional, not technical, and Section 3.4 already flags several; we restate the sharpest of them here as a research agenda rather than a design specification.

- At what transaction volume does automated micro licensing generate returns that exceed the cost of metering and dispute resolution, and below what volume does the model fail economically, regardless of its technical elegance?
- Does a data cooperative structure (Scholz, 2016) risk being captured by whichever subgroup

within a community is best organised or most fluent in the licensing system, and what governance safeguards, rotating representation, independent audit, exit rights, actually prevent this in practice rather than only on paper?

- How does revenue distribution work across borders and tax jurisdictions when a licensee, a community, and the infrastructure connecting them may sit in three different countries, and what compliance burden does that place on a community with no existing legal or accounting infrastructure of its own?
- Once initial grant or philanthropic funding for a pilot ends, does the architecture generate enough ongoing revenue to sustain itself, or does it quietly become dependent on renewed external funding in a way that undermines the self determination it is meant to support?

Cross Cutting: Methodology and Institutional Design

A final set of questions applies across all four pillars, and concerns how this programme should be run, not only what it should build.

- What does success look like here, measured not in technical benchmarks but in the kind of outcome Jack and Suri (2014) document for mobile money: a demonstrable, durable improvement in the economic position of the people the system was built for?
- Following the caution embedded throughout the Indigenous data governance literature (Christen, 2012; Carroll et al., 2020), how does a research programme convened at a university, and funded by external grants, avoid reproducing, procedurally, the very extraction it is technically designed to prevent?
- What does an appropriate evaluation timeline look like, given that community trust, economic outcomes and cultural fit are likely to take years to become visible, on a timescale that does not match a typical grant cycle or publication schedule?
- How much of this architecture generalises across the frontier and emerging markets named in Section 2, and how much needs to be redesigned from scratch for each new cultural and legal context, rather than assumed to transfer directly from the Saving Voices case?

7 Conclusion

The technologies to connect the world's remaining unserved communities now exist in early, separate forms. So do the technologies to let their cultural and linguistic heritage be heard, protected and economically sustained. Frugal AI can run where cloud infrastructure cannot reach. Space based connectivity and orbital compute can extend that reach further still. Indigenous data governance frameworks and federated learning together offer a technical path to privacy and cultural control that does not require centralising sensitive material. Bottom up data governance and platform cooperativism offer structural alternatives to extractive monetisation. What has been missing is not any single piece of this puzzle. It is the integration of all four into one coherent system, designed from the outset around frontier and emerging markets as the origin of genuinely new models, not their eventual destination. That integration is the research vision this paper sets out, building on the early evidence of the Saving Voices Initiative, and it is the programme of work this paper proposes to pursue.

References

- Abadade, Y., Temouden, A., Bamoumen, H., Benamar, N., Chtouki, Y., and Hafid, A.S. (2023). A Comprehensive Survey on TinyML. *IEEE Access*, 11, 96892–96922.
- AfriLabs (2024). *Unlocking Africa’s Potential for Sustainable Growth through the Creative Industry*. AfriLabs.
- Banerjee Belur, S., Sathiaselalan, N., Bannulmath, P., Saumya, S., Maralappanavar, S., Deepak, K.T., Prasanna, S.R.M., and Sathiaselalan, A. (2026). Frugal Voice: Offline Community-Governed Speech Models for the Soliga Tribe in Karnataka. In *Proceedings of the ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (COMPASS ’26)*, 447–456. New York: ACM. DOI: 10.1145/3811242.3819092.
- Bird, S. (2020). Decolonising Speech and Language Technology. In *Proceedings of the 28th International Conference on Computational Linguistics (COLING 2020)*, 3504–3519. Barcelona: International Committee on Computational Linguistics.
- Carroll, S.R., Garba, I., Figueroa-Rodríguez, O.L., Holbrook, J., Lovett, R., Materechera, S., Parsons, M., Raseroka, K., Rodriguez-Lonebear, D., Rowe, R., Sara, R., Walker, J.D., Anderson, J., and Hudson, M. (2020). The CARE Principles for Indigenous Data Governance. *Data Science Journal*, 19(1), 43.
- Christen, K.A. (2012). Does Information Really Want to be Free? Indigenous Knowledge Systems and the Question of Openness. *International Journal of Communication*, 6, 2870–2893.
- Couldry, N., and Mejias, U.A. (2019). *The Costs of Connection: How Data is Colonizing Human Life and Appropriating it for Capitalism*. Stanford, CA: Stanford University Press.
- Delacroix, S., and Lawrence, N.D. (2019). Bottom-up Data Trusts: Disturbing the “One Size Fits All” Approach to Data Governance. *International Data Privacy Law*, 9(4), 236–252.
- Denby, B., and Lucia, B. (2020). Orbital Edge Computing: Nanosatellite Constellations as a New Class of Computer System. In *Proceedings of the Twenty-Fifth International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS ’20)*, 939–954. New York: ACM.
- Frugal AI Hub, Centre for India & Global Business (2025). *Accelerating a Frugal AI Ecosystem* (White Paper). Cambridge: Cambridge Judge Business School, University of Cambridge.
- Gwaka, L., Haseki, M., and Yoo, C.S. (2023). Community Networks as Models to Address Connectivity Gaps in Underserved Communities. *Information Development*.
- Han, H., and Siebert, J. (2020). TinyML: A Systematic Review and Synthesis of Existing Research. *IEEE Circuits and Systems Magazine*, 20, 4–18.
- International Telecommunication Union (2025). *Facts and Figures 2025: Measuring Digital Development*. Geneva: ITU.
- Jack, W., and Suri, T. (2014). Risk Sharing and Transactions Costs: Evidence from Kenya’s Mobile Money Revolution. *American Economic Review*, 104(1), 183–223.
- McMahan, H.B., Moore, E., Ramage, D., Hampson, S., and Agüera y Arcas, B. (2017). Communication-Efficient Learning of Deep Networks from Decentralized Data. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR 54, 1273–1282.
- Moseley, C. (ed.) (2010). *Atlas of the World’s Languages in Danger*, 3rd edition. Paris: UNESCO Publishing.
- Prabhu, J., Narayan, P., and Sud, M. (2026). *LeanSpark: Frugal by Design, Global in Impact*. Gurugram: Penguin Business, Penguin Random House India.
- Radjou, N., and Prabhu, J. (2015). *Frugal Innovation: How to Do More with Less*. London: Profile Books.
- Radjou, N., Prabhu, J., and Ahuja, S. (2012). *Jugaad Innovation: Think Frugal, Be Flexible, Generate Breakthrough Growth*. San Francisco: Jossey-Bass.
- Saldana, J. (ed.), Arcia-Moret, A., Braem, B., Pietrosevoli, E., Sathiaselalan, A., and Zennaro, M. (2016). *Alternative Network Deployments: Taxonomy, Characterization, Technologies, and Architectures*. RFC 7962. Internet Research Task Force.
- Scholz, T. (2016). *Platform Cooperativism: Challenging the Corporate Sharing Economy*. New York: Rosa Luxemburg Stiftung.
- Sathiaselalan, A. (2026). *Rethinking Multilingual Speech for Under-Represented Languages: Tokenisation,*

Training, and Evaluation Beyond the English-Centric Paradigm. Position Paper, Saving Voices Project. Available at: www.savingvoices.org/position-paper.pdf.

Te Hiku Media (2021). *Kaitiakitanga License*. Kaitaia, New Zealand: Te Hiku Media.

United Nations Conference on Trade and Development (2024). *Creative Economy Programme: Contributions of the Creative Economy*. Geneva: UNCTAD.